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Competition: NANO NAVIGATOR

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MULTIDIRECTIONAL SENSING IR POWERED MICROMOUSE

Abstract

This project describes the development of a high-density, half-size micromouse maze-solving robot while minimizing the footprint to 45.7mm x 40.0mm. The system's brain is an STM32F405RG microcontroller in an LQFP64 package, which exploits high-speed processing capability in managing a sophisticated sensor fusion suite comprising a VL6180X ToF sensor for frontal ranging and four SFH-4045N/SFH-3015-FA infrared pairs for wall tracking along its sides. Due to ambient light interference, the IR emitter intensities are dynamically controlled by a PWM-controlled N-channel power MOSFET circuit.

The mechanical drive system utilizes ultra-compact 6mm coreless motors integrated with a custom-engineered offset gearbox to overcome torque limitations while maintaining a sub-1mm 3D-printed mounting tolerance. High-precision odometry is achieved with AS5600 magnetic encoders and an MPU6500 6-axis IMU interfaced via high-speed SPI for real-time orientation feedback. Power management is provided by a 3.7V LiPo battery regulated through a 1A low-dropout (LDO) linear regulator, with a DRV8833 dual H-bridge driver supplying up to 1.5A per channel for motor actuation. This architecture represents a significant achievement in spatial optimization, housing over 90 components on a custom-routed PCB to create one of the most compact STM32-based micromouse platforms.

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Design Overview

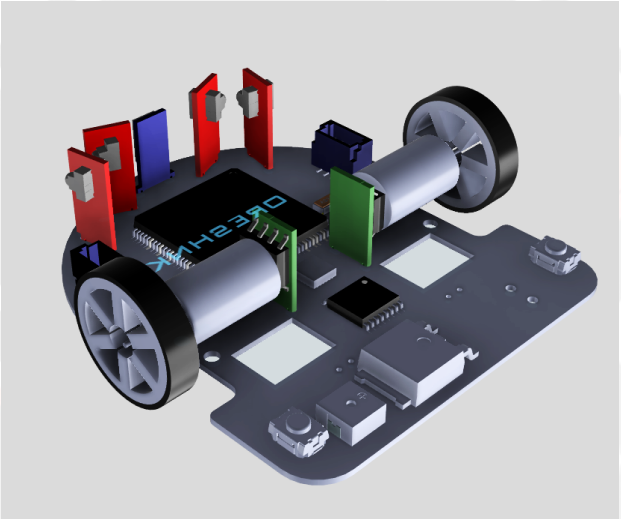

Fig 1.1 - This features the CAD design of the micromouse

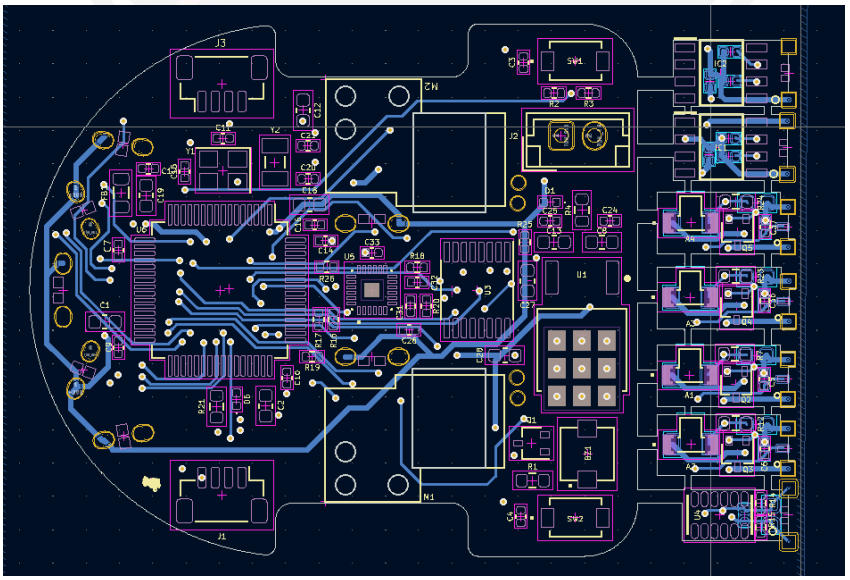

Fig 1.2 - PCB Design of the micromouse

Furthermore, the design principle used for the design of this bot includes an ultra-compact and highly dense printed circuit board architecture based on the half-size micromouse standard. In terms of navigation, the navigation suite uses a hybrid sensor set with a frontal VL6180X Time-of-Flight sensor and four IR emitter/receiver pairs angled to provide maximum spatial awareness.

To optimize mechanical efficiency within the specified footprint of 45.7mm x 40.0mm, the bot utilizes N20 motors in conjunction. This inventive gearbox mounting technique bypasses the torque constraints of coreless motors, thus facilitating the utilization of 3D housing with tolerances smaller than 1mm in size. Additionally, the utilization of AS5600 magnetic encoders alongside an MPU6500-based 6-axis inertial measurement unit guarantees ideal odometry and orientation feedback for the bot, thus ensuring high velocity stability coupled with pinpoint accuracy during sharp turns and diagonal movements.

Compliances in design

The bot meets all the necessary compliances including that of sizes.

<u>Driving Mechanism</u>

The type of motion system incorporated by the bot utilizes a highly efficient differential drive system with a DRV8833 dual H-bridge motor driver. Unlike other motor drivers, the DRV8833 motor driver integrates low-resistance MOSFETs that offer very low voltage drops, thus ensuring the delivery of maximum voltage from the 3.7V LiPo battery source. The system uses 2 six-millimeter coreless motors with a custom-ratio gearbox that provides high RPM with high torque to enable quick acceleration for half-size maze racing.

Stability and agility are achieved through a special mechanical setup:

Traction and Odometry: The bot uses high-grip mini-tires that are coupled with AS5600 magnetic encoder technology. The encoder technology is needed to perform precise maneuvers, like the high-speed 90-degree rotation and diagonal motion.

Low Friction Pivot: To address any mechanical drag that may result during fast turns due to high velocity, the chassis utilizes a precise ball caster balance. The precision ball caster acts as a frictionless pivot to avoid any parasitic drag from the mass of the battery pack, as well as the dense PCB.

Dynamic Motion Control: By utilizing complex Pulse Width Modulation (PWM) via the STM32F405RG, the bot can be fine-tuned on its path of motion. It can thus change motor speeds in real-time, thereby keeping up with the racing path with utmost accuracy.

Sensors used

Fig - All the sensors used in making the micromouse

The sensor architecture utilizes a high-performance hybrid array designed for the extreme spatial constraints of the half-size micromouse standard. In place of a traditional linear array, the bot employs a strategic configuration of four IR emitter-receiver pairs (SFH-4045N and SFH-3015-FA) integrated with a VL6180X Time-of-Flight (ToF) sensor. This multi-layered sensing approach allows the STM32F405RG to execute high-frequency PID corrections, enabling smoother trajectory transitions and higher maintainable velocities during complex maze navigation.

Motors Used

Fig - N20 motors and compatible RD-34 Wheels

We have used GA12-N20 6V 2000RPM Micro DC Metal Gear Reduction Motor for high speed movement of the bot.

<u>Complete List of Materials Used</u>

1) DRV8833PWPR
2) VL6180V1NR/1
3) MPU-6500
4) SFH 4045N
5) SmartElex HC-08 Bluetooth 4.0 BLE Serial Port Module
6) AS5600
7) APA3010P3BT-GX
8) Delivery - DigiKey
9) STM32F405RGT6
10) WLY602025 3.7V 250mAh 1S LiPo Battery
11) CPF0805F10KC1
12) CL21B104KBCNNNC
13) CL21A106KPFNNNE
14) PTS645SM50SMTR92 LFS
15) ST-LINK/V2 in-circuit debugger/programmer for STM8 and STM32
16) Ball Caster
17) GA12-N20 6V 2000RPM Micro DC Metal Gear Reduction Motor
18) AMS1117 3.3V Power Supply Module

Conclusion

We have created a micromouse which is a fast maze solving microbot , we took reference from Project Futura . We have worked upon our selection of components as compared to the material we have taken reference form , also we have reduced some redundancies on the PCB design , which resulted in better signaling. Unlike the reference material we have used N20 Motors with 2000rpm for higher speeds and also have attached ball caster in front for better bankings at sharp turns.